

Assignment Report

Delta Lake Incremental Data Processing

Objective

The objective of this assignment was to learn how Delta Lake handles incremental data processing using the MERGE operation in PySpark. The assignment demonstrates updating existing records and inserting new records into a Delta table.

Dataset Description

Master Dataset

The Sample Superstore dataset was used as the master dataset. It contains retail sales information such as customer details, product details, order information, sales, quantity, discount, and profit.

Incremental Dataset

A separate incremental dataset was created to simulate new incoming data. It includes:

- Existing records with modified values for updates.
- New records with unique IDs for inserts.

This simulates how data is incrementally added in real-world data engineering pipelines.

Steps Performed

1. Loaded the master dataset into PySpark.
2. Checked for null values and duplicate records.
3. Renamed columns for Delta compatibility.
4. Stored the dataset in Delta format.
5. Loaded the incremental dataset.
6. Matched its schema with the Delta table.
7. Applied the Delta Lake MERGE operation.
8. Verified the final output.

Why These Steps?

- Data quality checks ensured clean input data.
- Column names were standardized to avoid compatibility issues.
- The incremental dataset was prepared with matching schema so it could be merged correctly.
- MERGE was used to perform updates and inserts in a single operation.

Note

This assignment was developed and executed on **Databricks Free Edition**.

Before uploading to GitHub, most `display()` statements were removed because their outputs increased the notebook size and caused rendering issues on GitHub. The notebook logic remains unchanged and can be executed in Databricks to reproduce the results.

Results

The merge operation completed successfully by:

- Updating existing records.
- Inserting new records.
- Maintaining a consistent Delta table.

Conclusion

This assignment demonstrated how Delta Lake simplifies incremental data processing by handling updates and inserts through a single MERGE operation, making data pipelines more efficient and reliable.